

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60

Minutes Annexure A

Scenario-Based Questions

Code of Conduct:

I S.Varsha Reddy(192572177)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:S.Varsha Reddy

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	

Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
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Scenario-Based Questions

1.Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

- (a) Clearly interpret the scenario and explain the cause of low contrast.**
- (b) Identify all key issues affecting image visibility.**
- (c) Apply an appropriate enhancement technique to address the problem.**
- (d) Justify your choice with logical reasoning.**
- (e) Present your explanation in a clear and structured manner.**

Ans:

Low Contrast Image

(a) Scenario Understanding and Cause of Low Contrast

A grayscale image with low contrast appears dull because the intensity values of objects and the background are very close to each other. This reduces the difference between bright and dark regions, making objects difficult to distinguish.

Cause of Low Contrast:

- Pixel intensity values are concentrated within a narrow range.
- Poor lighting conditions during image acquisition.
- Insufficient difference between object intensity and background intensity.
- Limited dynamic range of the image.

(b) Identification of Key Issues Affecting Image Visibility

The major issues affecting visibility are:

1. Poor separation between object and background

- Objects cannot be clearly distinguished from surrounding regions.

2. Limited intensity range

- The grayscale values occupy only a small portion of the available range (0–255).

3. Reduced image details

- Fine features and edges become difficult to identify.

4. Low brightness variation

- The image lacks clear dark and bright regions.

5. Difficulty in image analysis

- Computer vision algorithms may struggle to detect objects accurately.

(c) Application of Enhancement Technique

Selected Technique: Histogram Equalization

Histogram equalization is applied to improve the contrast of the grayscale image.

Working Principle:

- The histogram of pixel intensities is redistributed.
- Pixel values are spread over the full grayscale range.
- Dark pixels become darker and bright pixels become brighter.
- This increases the difference between objects and background.

(d) Justification of the Technique

Histogram equalization is chosen because it is suitable for images where intensity values are concentrated in a narrow range. It automatically enhances contrast without requiring manual adjustment.

Reasons for Selection:

- Improves overall contrast effectively.
- Reveals hidden details in dark and bright areas.
- Enhances edges and object boundaries.
- Improves image quality for further processing tasks such as object detection and classification.

(e) Structured Explanation

A low contrast grayscale image occurs when object and background intensity values are very similar, making the image appear dull and unclear. The main problems include poor visibility, reduced details, and limited intensity distribution.

To solve this problem, histogram equalization is applied. This technique redistributes pixel intensity values across the available grayscale range, increasing contrast and making important features more visible.

After enhancement, the difference between objects and background increases, resulting in a clearer image suitable for human observation and computer vision applications.

2.Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpret the scenario and explain the impact of uneven illumination.

(b) Identify key factors causing intensity variation.

(c) Apply a suitable enhancement method to normalize illumination.

(d) Justify your selected approach with reasoning.

(e) Provide a clear and well-structured explanation.

Ans:

Uneven Illumination

Scenario Analysis

An image captured under non-uniform lighting conditions contains bright and dark patches because the amount of light falling on different regions of the image is not equal. This causes variations in pixel intensity even when the actual objects have the same surface properties.

Uneven illumination affects the visual quality of an image by creating false intensity differences, making some regions appear brighter or darker than they actually are.

Reasons for Uneven Illumination:

- Non-uniform lighting during image acquisition.
- Shadows created by objects or environmental conditions.
- Variations in light source intensity.
- Improper camera positioning or exposure settings.
- Reflection or absorption of light from object surfaces.

Problems Affecting Image Quality

The main issues caused by uneven illumination are:

False Intensity Variation

Regions of the same object may appear different due to lighting changes rather than actual differences.

Loss of Important Details

Dark areas may hide important features, while overly bright areas may lose information.

Difficulty in Object Detection

Computer vision systems may incorrectly identify objects because of brightness variations.

Reduced Image Uniformity

The image lacks consistent brightness, affecting analysis and interpretation.

Enhancement Technique Used

Background Normalization (Illumination Correction)

Background normalization is used to reduce the effect of uneven lighting by estimating and removing the illumination component from the image.

Process:

1. Estimate the illumination pattern using smoothing techniques.
2. Separate the illumination component from the original image.
3. Normalize the image intensity values.
4. Obtain a corrected image with uniform brightness.

Reason for Selecting Illumination Correction

Illumination correction is selected because the main problem is caused by variations in lighting rather than the actual image content. This method removes unwanted brightness changes while preserving important object details.

Advantages:

- Reduces bright and dark patches.
- Produces uniform illumination across the image.
- Improves visibility of hidden details.
- Enhances accuracy of image processing tasks such as segmentation and object recognition.
- Maintains important features and edges.

Conclusion

Uneven illumination causes unwanted intensity variations in an image due to non-uniform lighting conditions. These variations reduce image quality, hide important details, and affect the performance of computer vision algorithms.

3.Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

- Explain the scenario and characteristics of this noise.**
- Identify key issues impacting image quality.**
- Apply an appropriate filtering technique.**
- Justify why this method is more suitable than alternatives.**
- Present your response clearly and logically.**

Ans:

Salt-and-Pepper Noise

Scenario Analysis

A scanned document affected by salt-and-pepper noise contains random black and white pixels that appear scattered throughout the image. This type of noise reduces the clarity of text and makes the document difficult to read.

Salt-and-pepper noise is commonly caused by errors during image acquisition, transmission problems, or sensor defects. It appears as sudden intensity variations where some pixels become completely black (pepper noise) or completely white (salt noise).

Characteristics of Salt-and-Pepper Noise:

- Appears as random black and white spots.
- Affects individual pixels without changing the surrounding pixels.
- Produces sudden intensity changes.
- Reduces image clarity and readability.
- Commonly occurs in scanned documents and digital images.

Problems Affecting Image Quality

The main issues caused by salt-and-pepper noise are:

Reduced Text Readability

Noise spots overlap with characters, making letters unclear and difficult to recognize.

Loss of Image Details

Important features such as edges and fine structures may become distorted.

False Pixel Information

Random black and white pixels introduce incorrect intensity values into the image.

Difficulty in Image Processing

Noise can reduce the accuracy of OCR (Optical Character Recognition) and other computer vision applications.

Enhancement Technique Used

Median Filter

The median filter is applied to remove salt-and-pepper noise while preserving important edges and details.

Process:

1. Select a neighborhood window around each pixel (for example, 3×3).
2. Arrange the pixel values in ascending order.
3. Replace the center pixel with the median value.
4. Repeat the process for all pixels in the image.

Reason for Selecting Median Filter

The median filter is selected because salt-and-pepper noise consists of extreme pixel values (0 and 255). The median operation removes these unwanted extreme values without significantly affecting the actual image details.

Advantages:

- Effectively removes salt-and-pepper noise.
- Preserves edges better than averaging filters.
- Prevents excessive image blurring.
- Maintains text clarity in scanned documents.
- Improves the performance of OCR and document analysis systems.

Comparison with Other Methods

Mean Filter

- Replaces pixels with average values.
- Reduces noise but causes blurring.
- Not suitable for preserving text edges.

Gaussian Filter

- Smooths noise using weighted averaging.
- May remove fine details along with noise.

Median Filter

- Removes noise while preserving edges.
- More suitable for impulse noise such as salt-and-pepper noise.

Conclusion

Salt-and-pepper noise introduces random black and white pixels that reduce the readability and quality of scanned documents. It affects text recognition, image details, and further image processing operations.

The median filter is the most suitable technique because it removes unwanted noise pixels while preserving important edges and text details. Therefore, it improves document clarity and makes the image more suitable for analysis and recognition applications.

4.Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

- Interpret the scenario and explain why edges are blurred.**
- Identify key issues related to smoothing and edge loss.**
- Apply an appropriate technique to restore edges.**
- Justify your decision with clear reasoning.**
- Structure your explanation clearly.**

Ans:Blurred Edges

Scenario Analysis

An image processed for noise removal appears overly smooth because excessive filtering has reduced the intensity differences between neighboring pixels. As a result, important edges and fine details become less visible.

Edges represent the boundaries between different objects in an image. When strong smoothing filters are applied, they remove not only noise but also important high-frequency information, causing edges to become blurred.

Reasons for Edge Blurring:

- Excessive use of smoothing filters.
- Large filter window size during noise reduction.
- Removal of high-frequency components containing edge information.
- Over-smoothing of image details.
- Improper selection of filtering parameters.

Problems Affecting Image Quality

The main issues caused by blurred edges are:

Loss of Edge Information

Object boundaries become unclear, making it difficult to identify shapes and structures.

Reduced Image Sharpness

The image appears soft and lacks fine details.

Loss of Important Features

Small structures and textures may disappear during excessive smoothing.

Difficulty in Object Detection

Computer vision algorithms may struggle to locate and recognize objects accurately due to unclear boundaries.

Enhancement Technique Used

Laplacian Sharpening

Laplacian sharpening is applied to restore edge details by emphasizing rapid changes in pixel intensity.

Process:

1. Apply a Laplacian filter to detect edges and intensity changes.
2. Combine the detected edge information with the original image.
3. Enhance high-frequency components to make boundaries sharper.
4. Obtain an image with improved edge visibility.

Reason for Selecting Laplacian Sharpening

Laplacian sharpening is selected because the problem is caused by the loss of edge information after excessive smoothing. The Laplacian operator highlights intensity changes and restores the sharpness of object boundaries.

Advantages:

- Enhances edges and fine details.
- Improves image sharpness.
- Restores important high-frequency information.
- Makes object boundaries clearer.
- Improves performance of image analysis tasks such as segmentation and recognition.

Comparison with Other Methods

Increasing Contrast

- Improves brightness differences but does not effectively restore lost edges.

Histogram Equalization

- Enhances overall contrast but cannot recover blurred boundaries.

Laplacian Sharpening

- Directly enhances edges and fine details.
- More suitable for correcting over-smoothed images.

Conclusion

Over-smoothing during noise removal can remove important edge information along with noise, resulting in blurred boundaries and reduced image quality. Applying Laplacian sharpening helps restore edge details by enhancing intensity variations around object boundaries.

This technique improves image sharpness, preserves important features, and makes the image more suitable for computer vision applications such as object detection and image analysis.

5.High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

- (a) Interpret the scenario and describe high-frequency noise.**
- (b) Identify the key issues affecting image clarity.**
- (c) Apply an appropriate frequency domain filtering method.**
- (d) Justify the choice of filter with reasoning.**
- (e) Present your answer in a clear and organized manner.**

Ans:

High-Frequency Noise

Scenario Analysis

A satellite image affected by high-frequency noise contains small, rapid variations in pixel intensity that appear as fine-grained patterns or random disturbances. This type of noise reduces the clarity of important features such as land boundaries, roads, and structures.

High-frequency noise occurs due to sensor limitations, transmission errors, or environmental interference during image acquisition. It mainly affects the detailed components of an image and can make important features difficult to identify.

Characteristics of High-Frequency Noise:

- Appears as fine-grained or random intensity variations.
- Mainly affects edges and detailed regions of an image.
- Creates unwanted sharp changes in pixel values.
- Reduces image smoothness and visual quality.
- Commonly occurs in satellite and remote sensing images.

Problems Affecting Image Quality

The main issues caused by high-frequency noise are:

Reduced Feature Visibility

Important details such as terrain boundaries, buildings, and surface patterns become difficult to distinguish.

False Edge Detection

Noise may create unwanted edges, affecting accurate feature extraction.

Reduced Image Clarity

The image appears grainy due to unwanted high-frequency components.

Difficulty in Image Analysis

Noise can reduce the accuracy of classification, segmentation, and object detection algorithms.

Enhancement Technique Used

Low-Pass Frequency Domain Filtering

A low-pass filter is applied in the frequency domain to reduce high-frequency noise while preserving important low-frequency image information.

Process:

1. Convert the image from the spatial domain to the frequency domain using Fourier Transform.
2. Apply a low-pass filter to remove unwanted high-frequency components.
3. Suppress noise frequencies while allowing low-frequency components to pass.
4. Convert the filtered image back to the spatial domain using Inverse Fourier Transform.

Reason for Selecting Low-Pass Filtering

A low-pass filter is selected because high-frequency noise appears as rapid intensity changes in the frequency domain. By reducing high-frequency components, the filter removes noise and produces a smoother image.

Advantages:

- Effectively reduces fine-grained noise.
- Improves overall image smoothness.
- Preserves important large-scale structures.
- Enhances visibility of satellite image features.
- Improves accuracy of further image processing tasks.

Comparison with Other Methods

High-Pass Filtering

- Enhances high-frequency components such as edges.
- May increase noise instead of removing it.

Median Filtering

- Works well for salt-and-pepper noise.
- Less effective for frequency-based noise patterns.

Low-Pass Filtering

- Specifically reduces high-frequency variations.
- More suitable for removing fine-grained noise from satellite images.

Conclusion

High-frequency noise introduces unwanted fine variations in satellite images, reducing feature visibility and affecting image analysis. Applying low-pass frequency domain filtering effectively suppresses high-frequency components while preserving important image information.

This improves image clarity, reduces noise, and makes the satellite image more suitable for applications such as remote sensing, feature extraction, and image classification.